UNIVERSITY OF BUEA
FACULTY OF SCIENCE
FIRST SEMESTER EXAMINATION 2022/2023

DEPTARTMENT: COMPUTER SCIENCE	COURSE INSTRUCTOR(S): Denis Nkweteyim
MONTH/YEAR: June 2023	COURSE CODE: DSC608
DATE: 23/06/2023	COURSE TITLE: Information Retrieval
TIME ALLOWED: 3 Hours (08:00-11:00)	
CREDIT VALUE: 6	Venue: U-BLOCK D
INSTRUCTIONS: Answer All Questions	

Question 1: Zipf's Law and Heaps's Law ([2+2+2] + [2+2+4] = 14 marks)

a)

i) State Zipf's law, and write an expression to represent it.
ii) Sketch a graph to illustrate what Zipf's law represent?
iii) What are the limitations of Zipf's law (i.e., under what circumstances does the law ~~apply~~ not apply)?

b) The expression for Heaps' law is $v = k \cdot n^{\beta}$.

i) What do the parameters in Heaps' Law represent?
ii) Under what conditions does Heaps' Law apply?
iii) Consider a collection with one billion tokens (i.e., with 10^9 tokens). Suppose the first 1,000 of these tokens results in a vocabulary size of 1,000 terms, and the first 100,000 tokens results in a vocabulary size of 10,000 terms. Use Heap's law to estimate the vocabulary size of the whole collection. You may the table on the right useful.

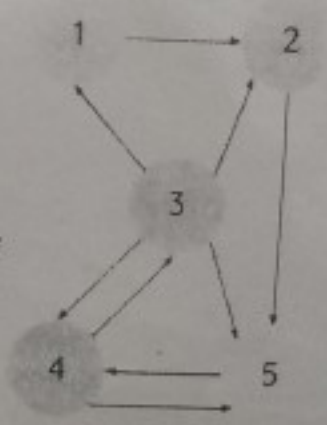
Logarithmic laws

Products:	$\log_b mn = \log_b m + \log_b n$
Ratios:	$\log_b \frac{m}{n} = \log_b m - \log_b n$
Powers:	$\log_b n^p = p \log_b n$
Roots:	$\log_b \sqrt[q]{n} = \frac{1}{q} \log_b n$
Change of bases:	$\log_b n = \log_a n \log_b a$

Question 2: Link Analysis and Data Structures ([3+3+8] + 6 = 20 marks)

a)

i) How does the PageRank algorithm classify hyperlinks for the purpose of determining the relative importance of web pages?
ii) Write an expression for the rank of a page using the PageRank algorithm. Clarify what the parameters stand for.
iii) Given the graph on the right, compute the page rank of each of the pages after two iterations of the page rank algorithm. Assume that the starting page rank of each of the pages is 1/5.



b) An incidence matrix and an inverted matrix are two candidates for storing indexing information in IR. With the aid of an example, explain what these data structures are. Compare their costs in terms of space complexity.

Question 3: Vector space model, inverted indexes, and ranking ([6+6] + [6+6] + [4+2+6] = 36 marks)

a) Suppose you wish to find reports regarding the impact of technology training on Cameroon's economy. You have a collection of three reports (A, B, and C). The table below shows how often some key words appear in each report.

	Technology	Training	Cameroon	Economy
Report A	12	0	3	24
Report B	10	5	20	10
Report C	0	12	9	8
Query	1	1	1	1

Your task is to rank the three reports using the cosine similarity measure for the query **Technology Training Cameroon Economy**.

i) Calculate the cosine similarity scores between the query and each document (you should ignore the **idf** term). Use the computed scores to rank the documents in order of relevance. Show your workings.

ii) Repeat the similarity computations and ranking you did above, this time making use of the **idf** term such that term weighs are now computed as $w_{i,j} = tf_{i,j} \times log(\frac{N}{df_i})$, where $\mathbf{tf_{i,j}}$ is term frequency of term **i** in document **j**; $\mathbf{df_i}$ is the document frequency of term **i**, and **N** is the total number of documents. Logarithms are taken to base 2.

Note: The general formula for calculating the cosine of the angle between two 4-dimensional vectors (x1, x2, x3, x4) and (y1, y2, y3, y4) is:

$$\cos(\vec{x}, \vec{y}) = \frac{x_1y_1 + x_2y_2 + x_3y_3 + x_4y_4}{\sqrt{x_1^2 + x_2^2 + x_3^2 + x_4^2}\sqrt{y_1^2 + y_2^2 + y_3^2 + y_4^2}}$$

b) Consider the following set of documents:

D1 : Do you like green eggs and ham
D2 : I do not like them Sam I am
D3 : I do not like green eggs and ham
D4 : I do not like eggs
D5 : Why are they green

Assuming there are no stop words, draw 2 inverted indexes for the document set as follows:

i) The first inverted index should include information on counts (i.e., document frequencies in the dictionary node and term frequencies in the postings list nodes)

ii) The second inverted index should include information on word positions (i.e., for each document, the position in which the term is found)

c) Now, consider that our IR schemes works as follows. For a query (e.g., **rising interest rate**), the system searches for the phrase **rising interest rate**, and if less than a threshold of **K** documents are retrieved, the system runs the following 2-phrase queries: **rising interest** and **interest rates**. If still, less than a total of **K** documents are retrieved, the system then runs one-word queries using each of the three words.

For our document collection in (b) above, let the query be **green eggs** and suppose we eventually want to show the user 4 documents (i.e., **K = 4**).

i) What are the documents that get retrieved in this first step? Illustrate how you arrived at your answer.

ii) What are the scores of the documents you retrieved in (I) above for the phrase query using cosine similarity (ignore the **idf** term)?

iii) If you did not have **K** documents in the first step, continue scoring according to the technique used in our IR scheme. What are the final 4 documents that you return and their scores)?